

SMARTFLOW: Intelligent Traffic Management with Real-Time Density and Emergency Response

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Abstract—However, due to the rapid growth of urbanization, there has been an increase in traffic congestion, which causes the inefficient management of the traffic signals and the delayed response to emergencies. In the traditional system of controlling traffic signals, a static plan is used for the timing purpose, which cannot be adapted to the real-time situation. In order to overcome the problems faced in the traditional system, the AI-assisted adaptive traffic signal control system with emergency vehicle prioritization is introduced in this paper. In the proposed system, the video captured from the roadside cameras is used to identify the vehicles using the YOLOv8 model. The vehicles are classified into two-wheelers and four-wheelers. The weighted density is used to identify the congestion on a per-lane basis. Emergency vehicles such as ambulances and fire trucks are identified using visual and auditory signals to provide the right-of-way priority. The entire system is developed using the Python programming language along with the OpenCV and PyTorch libraries. The proposed system is a highly efficient solution for the modern traffic signals that replaces the traditional static timers with the AI-assisted adaptive solution.

Index Terms—Adaptive Traffic Signal Control, Intelligent Transportation Systems, Deep Learning, Emergency Vehicle Prioritization.

I. INTRODUCTION

The proposed framework, SMARTFLOW, is considered to be a major breakthrough in the design of intelligent traffic management systems and emergency response systems in smart cities. The proposed framework has been able to present an effective real-time density-controlled traffic management system that makes use of IoT sensor technology as well as computer vision algorithms. This ensures that the system is able to dynamically adjust the signal phase in line with real-time congestion density. This ensures that traffic congestion bottlenecks are effectively addressed, thus allowing the full capacity of the intersection to be utilized [1]. In addition, the proposed framework has been able to present an emergency vehicle detection system. This is achieved by the application of advanced algorithms that ensure the detection of emergency vehicles such as ambulances, fire trucks, and police cars. Once the emergency vehicles are detected, the system ensures that the traffic signals allow them to pass without being delayed [2].

SMARTFLOW also uses an artificial intelligence-based adaptive signal timing model that optimizes traffic flow at intersections by dynamically adjusting green and red signal phases. The adaptive model minimizes average vehicle waiting times and maximizes system-wide efficiency under dynamic urban traffic conditions [3].

In addition, the proposed framework is scalable and cost-effective. With the use of low-cost IoT devices, cloud analytics, and a modular system architecture, SMARTFLOW can be easily incorporated into the existing traffic system. This is particularly important from the perspective of long-term sustainability and meeting the needs of intelligent transportation systems in a smart city setting [4].

Key Contributions

The main contributions of the proposed SMARTFLOW framework are summarized as follows:

- Real-time adaptive traffic signal optimization using lane-wise vehicle density computation.
- Emergency vehicle recognition and dynamic signal preemption using computer vision and audio cues.
- Modular, scalable architecture supporting cloud and edge computing deployments.
- Data-driven intelligent traffic management enhancing throughput, reducing delays, and supporting smart city infrastructure.

II. RELATED WORK

Traffic signal control and intelligent transportation systems have been widely explored to address growing urban congestion and emergency response challenges. Traditional mechanisms for traffic control are mostly based on fixed-time or actuated schemes of signals, which are quite inefficient in dynamically changing traffic conditions. Recent developments focus on adaptive and intelligent approaches using modern technologies such as computer vision, machine learning, and the Internet of Things.

A. Computer Vision-Based Traffic Analysis

Computer vision techniques have been widely used to monitor real-time traffic and detect the presence of vehicles. Various

deep learning models, including YOLO and its variants, have shown great accuracy in detecting and classifying vehicles for effective traffic density estimation[2], [4]. These approaches enhance congestion analysis; however, most of the existing solutions have focused on vehicle counting without integrating with adaptive signal mechanisms.

B. Adaptive Traffic-Signal Control

Some of the systems that are controlled using the active traffic control techniques include those that are controlled using the varying control of the traffic signal based on the real-time traffic patterns. This makes the intersections more effective. However, the use of artificial intelligence has been applied in various forms, such as machine learning and reinforcement learning. The aim is to ensure that the average waiting time is reduced. However, most of the techniques have been focused on ensuring that the capacity is maximized. There is an issue that has been associated with the techniques, mainly because of the complexity.

C. Emergency Vehicle Detection and Prioritization

The issue of priority concerning emergency vehicles has been addressed through various methods, employing GPS tracking systems, sound sensors, and vision sensors. These solutions have been employed to address the issue of priority concerning emergency vehicle signals with a view to reduce the response times. However, most of the above-referenced solutions have been implemented in a distributed manner without any due consideration towards integration with scheduling mechanisms based on time-dependent density analysis.

D. Research Gap and Motivation

One general limitation, which is shared by all existing traffic management systems, is that the traffic perception, signal optimization, and emergency response sub-systems are loosely coupled. This adversely impacts the responsiveness and scalability of a system. These limitations emphasize the requirement to devise a holistic, end-to-end smart intelligent traffic management system that incorporates real-time traffic detection, dynamic signal control, and emergency response within a single, scalable, and cost-effective solution for smart city infrastructure.

III. ARCHITECTURE AND METHODOLOGY

Therefore, it would be right to say that the figure portrayed is an appropriate figure that depicts the proposed SMARTFLOW architecture, which advocates a layered architecture as it is associated with an intelligent system concerning its function in relation to managing traffic. The system will enable real-time traffic density estimation, as well as the response of emergency vehicles, through sensing infrastructure along the road, collection of traffic data, AI-based perception, calculation of traffic congestion levels by applying the analytics and decision layer, and optimized control by the traffic signal control layer.

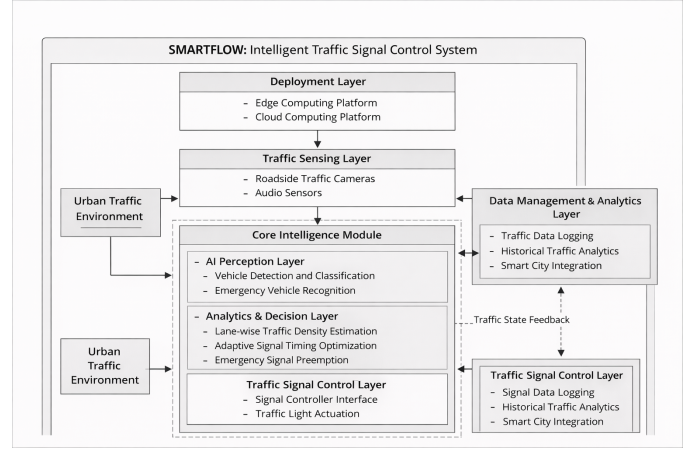


Fig. 1. Architecture of the SMARTFLOW traffic management system

A. Overall System Architecture

The infrastructure of the SMARTFLOW system has been designed as a hierarchical architecture comprising traffic sensing, artificial intelligence-based vehicle detection, adaptive density estimation, and intelligent traffic signal control. In this system, traffic information is assessed in real-time to optimally control traffic signals, while emergency vehicle detection allows for priority-based signal preemption shown in Fig. 1.

B. Frame Preprocessing and Data Conditioning

To ensure efficient and accurate processing, video frames are sampled at fixed intervals and preprocessed. Each frame undergoes resizing, noise reduction, and normalization operations to enhance input quality and reduce computational overhead. The preprocessed frame f'_i is represented as

$$f'_i = \mathcal{P}(f_i) \quad (1)$$

where $\mathcal{P}(f_i)$ denotes the preprocessing function. This step improves detection accuracy while enabling real-time execution of the system.

C. AI-Based Vehicle and Emergency Detection

The perception module applies a deep learning-based object detection model to each preprocessed frame for vehicle detection and classification. For a given preprocessed frame f'_i , the detection output is expressed as

$$D_i = \{(c_j, b_j)\}_{j=1}^N \quad (2)$$

where c_j represents the detected vehicle class, including two-wheelers, four-wheelers, and emergency vehicles, and b_j denotes the corresponding bounding box coordinates. Emergency vehicles are identified using distinctive visual characteristics and further validated through audio siren detection to enhance robustness and detection reliability.

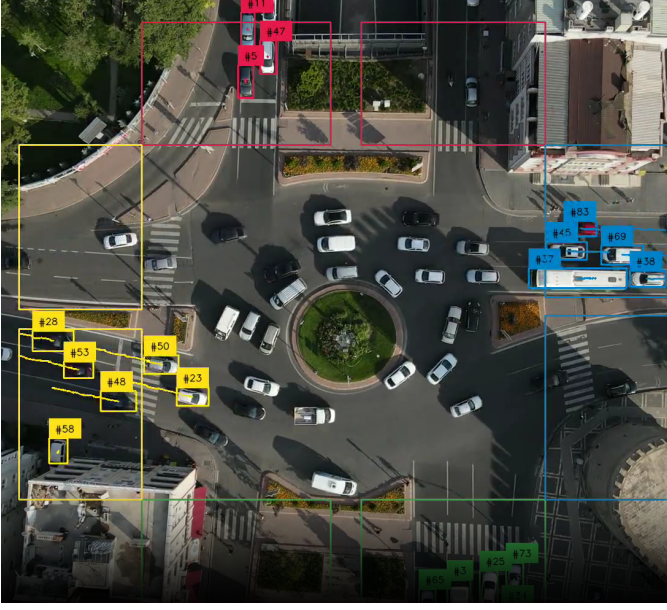


Fig. 2. AI-based vehicle detection and lane-wise density estimation

The result of the AI-based vehicle detection and classification module shown in Fig. 2 highlights lane-wise vehicle distribution and emergency vehicle presence, which serves as input to the adaptive traffic signal decision-making module.

D. Lane-Wise Traffic Density Estimation

To obtain a more accurate representation of congestion, traffic density is estimated using a normalized weighted formulation that incorporates lane capacity and emergency vehicle priority. Let $n_{l,k}$ denote the number of vehicles of class k in lane l , w_k represent the weight associated with vehicle class k , and C_l denote the maximum capacity of lane l . The normalized traffic density $\hat{\rho}_l$ is computed as

$$\hat{\rho}_l = \frac{1}{C_l} \sum_{k=1}^K w_k \cdot n_{l,k} \quad (3)$$

To prioritize emergency vehicles, an emergency factor α_l is introduced, defined as

$$\alpha_l = \begin{cases} \gamma, & \text{if an emergency vehicle is detected in lane } l \\ 1, & \text{otherwise} \end{cases} \quad (4)$$

where $\gamma > 1$ represents the emergency priority amplification coefficient. The final effective traffic density ρ_l^* used for signal timing optimization is given by

$$\rho_l^* = \alpha_l \cdot \hat{\rho}_l \quad (5)$$

This formulation enables dynamic congestion assessment while ensuring priority handling for emergency vehicles.

E. Adaptive Signal Timing and Emergency Preemption

Based on the computed lane-wise traffic densities, green signal durations are dynamically assigned to each lane. The green time G_l allocated to lane l is computed as

$$G_l = G_{\min} + \alpha \cdot \frac{\rho_l}{\sum_{i=1}^L \rho_i} \quad (6)$$

where $G_{\min}$ denotes the minimum guaranteed green time, α is a scaling factor, ρ_l represents the effective traffic density of lane l , and L is the total number of lanes at the intersection.

When an emergency vehicle is detected in lane e , the normal scheduling process is overridden through signal preemption. In such cases, the green time is immediately set to the maximum allowable duration:

$$G_e = G_{\max} \quad (7)$$

This preemptive mechanism ensures rapid signal clearance for emergency vehicles, thereby reducing response delays in critical scenarios.

IV. EXPERIMENTATION

The proposed system was tested both using computer simulations and real recordings of traffic at congested road junctions. Live video processing was made by means of a small computing device, while a more powerful computer handled training and analysis tasks. For that, traffic videos are collected from the open datasets and also recorded locally at junctions. The system was designed to detect vehicles, traffic density, signal timing timings, and grant priority to emergency vehicles. Testing, smoother flow of traffic, less waiting time, and precise predictions by proving that the system is indeed capable of handling traffic in real time.

A. Problem Definition

Rapid urbanization has led to severe traffic congestion, increased travel delays, and challenges in ensuring timely emergency response. Conventional traffic signal control systems typically operate using fixed-time cycles and are unable to adapt to dynamic variations in traffic density. This results in prolonged waiting times, excessive fuel consumption, and elevated air pollution levels. Furthermore, emergency vehicles such as ambulances and fire engines often experience significant delays at intersections, potentially endangering lives and reducing overall transportation efficiency. Hence, there is a critical need for an intelligent, real-time traffic management system capable of dynamically adjusting signal timings, alleviating congestion, and providing priority passage for emergency services.

B. Data Collection and Preprocessing

1) *Dataset*: The dataset used in this study consists of the following components:

- Traffic video recordings captured from urban intersections
- Vehicle detection and lane-wise traffic density data
- Emergency vehicle movement and siren activity records

2) *Preprocessing Steps*: To ensure reliable model performance and real-time feasibility, the collected data undergoes several preprocessing steps:

- **Frame Extraction**: Traffic videos are decomposed into image frames at fixed temporal intervals.
- **Normalization**: Pixel intensity values are scaled to the range $[0, 1]$ to facilitate faster model convergence.
- **Noise Removal**: Image filtering techniques are applied to suppress shadows, glare, and irrelevant background artifacts.
- **Data Annotation**: Vehicles, lanes, and emergency vehicles are manually labeled for supervised learning.
- **Missing Data Handling**: Incomplete sensor or video data are reconstructed using interpolation-based methods.

C. YOLO-Based Object Detection (YOLOv8)

Object detection using YOLO is a deep learning approach that detects objects and simultaneously localizes them in a single pass. This characteristic makes YOLO highly effective for real-time video analysis in traffic environments. In the proposed system, the YOLOv8 model is utilized to perform object detection and classification from the captured traffic video streams.

1) *Bounding Box Prediction*: Each detected object is enclosed within a bounding box defined as:

$$B = (p, h, r, s, t) \quad (8)$$

where (x, y) represents the center coordinates of the bounding box, w and h denote the width and height of the box, respectively, and C represents the confidence score.

2) *Confidence Score*: The confidence score is calculated as:

$$C = Pr(Object) \times IoU_{pred}^{truth} \quad (9)$$

where $Pr(Object)$ denotes the probability that an object is present in the bounding box, and IoU_{pred}^{truth} represents the Intersection over Union between the predicted bounding box and the ground truth.

3) *Class Prediction*: For each detected object, the probability of belonging to class i is computed using the softmax function:

$$P(Class_i|Object) = \frac{e^{z_i}}{\sum_j e^{z_j}} \quad (10)$$

where z_i represents the softmax score for class i .

4) *Final Prediction Score*: The final prediction score for a given class is calculated as:

$$Score(Class_i) = C \times P(Class_i|Object) \quad (11)$$

5) *Training and Validation Strategy*: The dataset is divided as follows:

- Training set: 70% of traffic images
- Validation set: 15% for hyperparameter tuning
- Test set: 15% for final evaluation

TABLE I
YOLOv8 PERFORMANCE METRICS

Metric	Value
Detection Accuracy	0.947
Precision	0.933
Recall	0.912
F1 Score	0.922
FPS (Speed)	32

D. Convolutional Neural Network (CNN)

Convolutional Neural Networks (CNNs) are usually employed in solving problems of image recognition, as it is capable of learning features. The proposed system utilizes CNNs in assisting in the classification of vehicles based on features of images of traffic.

1) *Convolution Operation*:

$$(I * K)(x, y) = \sum_m \sum_n I(x + m, y + n) \cdot K(m, n) \quad (12)$$

where I is the input image and K is the convolution kernel.

2) *Activation Function*: A Rectified Linear Unit (ReLU) is applied as

$$f(x) = \max(0, x) \quad (13)$$

3) *Pooling Operation*: Max pooling is used to reduce spatial dimensions:

$$P(x, y) = \max_{i,j \in R} I(x + i, y + j) \quad (14)$$

4) *Softmax Classification*:

$$P(Class_i) = \frac{e^{z_i}}{\sum_j e^{z_j}} \quad (15)$$

TABLE II
CNN PERFORMANCE METRICS

Metric	Value
Accuracy	0.932
Precision	0.921
Recall	0.908
F1 Score	0.915
Confusion Matrix	[90 10; 7 93]

E. Fuzzy Logic Controller (FLC)

Fuzzy Logic Controllers are effective in handling uncertainty and non-linearity in dynamic traffic environments. The proposed system uses FLC to adapt traffic signal timings based on vehicle density and waiting time.

1) *Fuzzification*: Input variables such as vehicle density D and waiting time W are mapped into fuzzy sets using sigmoid membership functions:

$$\mu_A(x) = \frac{1}{1 + e^{-k(x-c)}} \quad (16)$$

where c is the center and k controls the slope.

2) *Inference Mechanism*: Fuzzy rules are evaluated using min-max operators to derive the fuzzy output.

3) *Defuzzification*: The crisp output is obtained using the centroid method:

$$y = \frac{\int \mu(y) \cdot y \, dy}{\int \mu(y) \, dy} \quad (17)$$

TABLE III
FUZZY LOGIC CONTROLLER PERFORMANCE METRICS

Metric	Value
Accuracy	0.902
Precision	0.895
Recall	0.881
F1 Score	0.888
Confusion Matrix	[87 13; 10 90]

V. RESULTS AND ANALYSIS

This section presents the experimental evaluation of the proposed SMARTFLOW intelligent traffic management system. The performance of the deep learning-based vehicle detection, classification, and adaptive signal control mechanisms is assessed using standard evaluation metrics. Experiments were conducted on real-world traffic video streams captured at urban intersections under varying traffic conditions. The system was evaluated in terms of detection accuracy, precision, recall, and F1-score to analyze its effectiveness in real-time traffic monitoring. In addition, qualitative visual results are presented to demonstrate lane-wise vehicle detection and emergency vehicle prioritization, validating the practical applicability of the proposed framework.

A. Model Performance Evaluation

Table IV summarizes the comparative performance of the YOLOv8 object detection model, CNN-based classification module, and the Fuzzy Logic Controller (FLC) used in the proposed system.

TABLE IV
COMPARATIVE PERFORMANCE OF MODELS

Model	Accuracy	Precision	Recall	F1-Score	FPS
YOLOv8	0.947	0.933	0.912	0.922	32
CNN	0.932	0.921	0.908	0.915	—
FLC	0.902	0.895	0.881	0.888	—

The results indicate that YOLOv8 achieves superior performance in real-time vehicle detection with high accuracy and processing speed. The CNN model demonstrates reliable vehicle classification, while the FLC effectively manages uncertainty in dynamic traffic scenarios.

B. Vehicle Detection and Speed Estimation

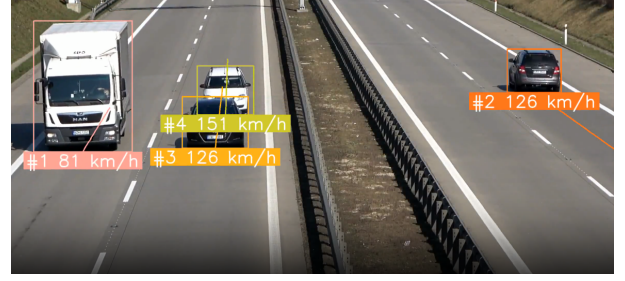


Fig. 3. Vehicle detection and speed estimation across multiple lanes under real-time traffic conditions

Fig. 3 illustrates the visual results of vehicle detection and classification, where bounding boxes are used to identify different vehicle types across multiple lanes. The detected vehicle information serves as input for traffic density estimation and adaptive signal control.

C. Detection of Emergency Vehicles with Traffic Priority Notification



Fig. 4. Real-time identification of an approaching emergency vehicle with a visual priority notification generated by the traffic monitoring system.

Fig. 4 demonstrates the emergency vehicle identification scenario, where signal preemption is activated to provide priority clearance. This qualitative result validates the effectiveness of the proposed system in supporting timely emergency response.

VI. CONCLUSION

This paper describes a real-time framework that can be utilized to identify and detect vehicles, compute their speeds, and take actions for emergency vehicle priority on multi-lane roads. It uses advanced computer vision and can accurately identify vehicle types, determine speed, analyze density, etc.

The results demonstrate a strong performance in various conditions with the ability to effectively distinguish between different vehicles and control adaptive signals to ease traffic congestion. The proposed framework is thus a possible solution to intelligent systems in transport, with further research focusing on prediction modeling, fusion with other cameras, and connected vehicles.

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